

Self-Supervised Learning based on Heat Equation

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❖ CAM shows that CNNs followed by GAP is able to learn *categorical heatmap*

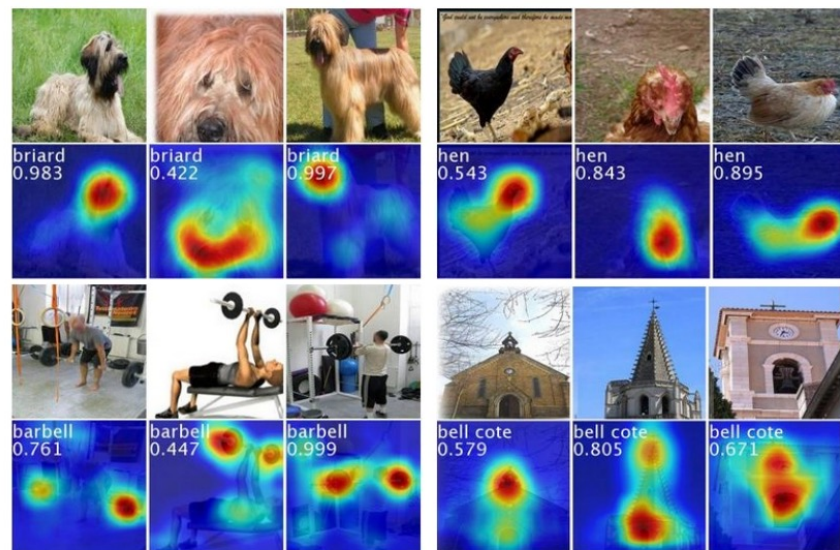
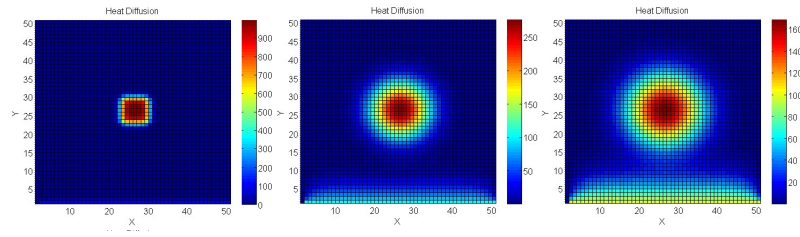
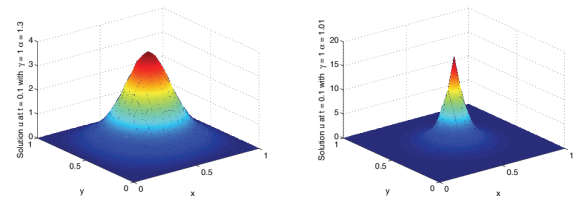
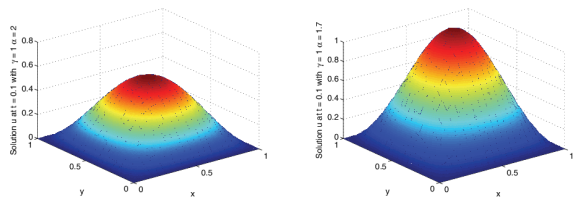


Figure 3. The CAMs of four classes from ILSVRC [20]. The maps highlight the discriminative image regions used for image classification e.g., the head of the animal for *briard* and *hen*, the plates in *barbell*, and the bell in *bell cote*.

❖ CAM is similar to physical heat diffusion



Heat equation

Brushing teeth



Cutting trees



CAM

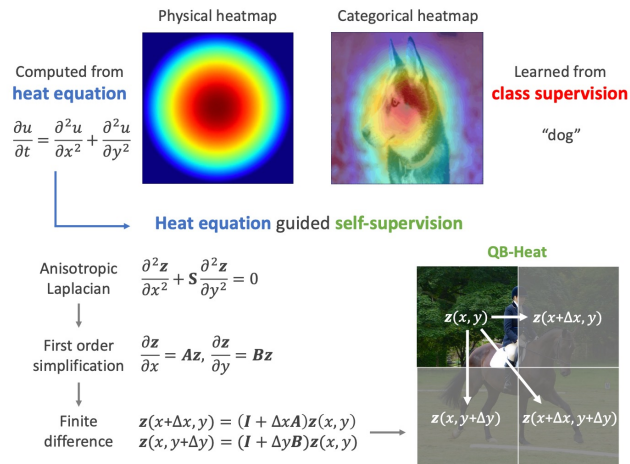
❖ Measurable scalar to latent vector

Add steady-state condition $\frac{\partial z}{\partial t} = 0$ to remove time dependence

Extend 2D isotropic Laplacian into linearly correlated

Anisotropic Laplacian $\frac{\partial^2 z}{\partial x^2} + S \frac{\partial^2 z}{\partial y^2} = 0$, where z is feature map with C channels, $z(x, y) \in \mathbb{R}^C$ and S is a $C \times C$

- S plays two roles:
 - handling nonequivalent change over horizontal and vertical directions
 - encoding invariant relationship between the second order of derivatives along x and y axes in the latent representation space



❖ Measurable scalar to latent vector

Add steady-state condition $\frac{\partial z}{\partial t} = 0$ to remove time dependence

Furthermore, decouple this spatial invariance along x and y axes separately to simplify the anisotropic Laplacian into two first order linear differential equations as follows:

First order linear:

$$\frac{\partial z}{\partial x} = \mathbf{A}z, \frac{\partial z}{\partial y} = \mathbf{B}z, \text{ where } \mathbf{A} \text{ and } \mathbf{B} \text{ are invertible matrices with size } C \times C \text{ and } S = -\mathbf{A}^2(\mathbf{B}^2)^{-1}$$

- Holding linear relationship for any order derivatives between x and y

Finite difference:

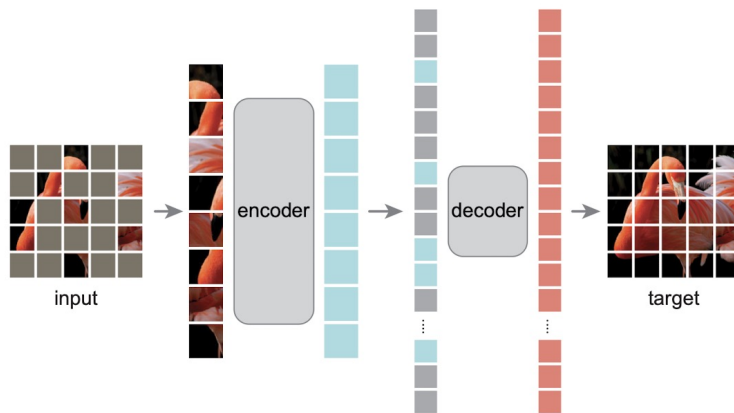
$$z(x + \Delta x, y) - z(x, y) = \Delta x \mathbf{A}z(x, y)$$

$$z(x, y + \Delta y) - z(x, y) = \Delta y \mathbf{B}z(x, y)$$

- This gives rise to a new masked image modeling method
- Specifically, only a single quarter-block is unmasked to encode $z(x, y)$, which is used to predict other three masked quarters $z(x + \Delta x, y)$, $z(x, y + \Delta y)$, $z(x + \Delta x, y + \Delta y)$ via linear prediction.

❖ Compared to popular MAE, it has four difference

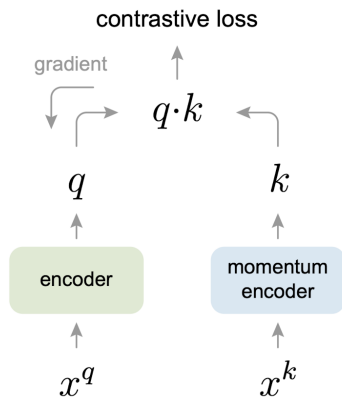
1. More regular masking
2. Simpler linear prediction
3. Enabling masked image modeling for efficient CNN
4. Modeling spatial invariance explicitly in representation space via learnable matrices A and B



❖ Representation learning method in SSL

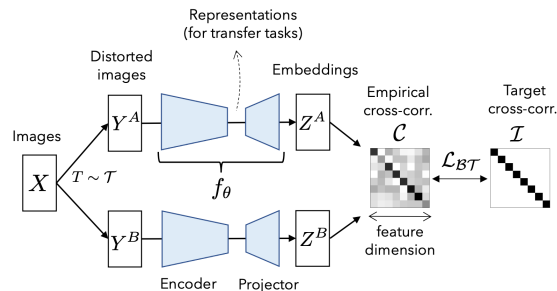
Contrastive learning

Ex) DINO, SimSiam, **MoCo**, CPC



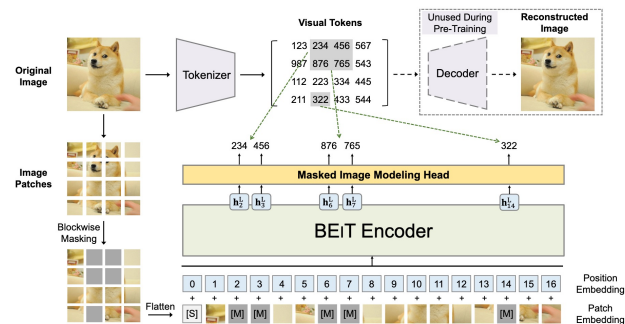
Information maximization

Ex) **Barlow Twins**, VICReg, W-MSE



Masked image modeling

Ex) **BEiT**, MaskFeat, MAE



❖ Heat equation in feature space

Uni-dimensional and observed variable into multi-dimensional and latent feature space

- How to extend heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ from u into latent feature space z

Motivation:

- Motivated by class activation maps (CAM) in which the categorical heatmap is similar to physical heat diffusion

Hypothesize:

- (a) the feature map around a visual object is smooth and governed by heat equation
- (b) the corresponding feature encoder can learn from heat equation alone without any labels

Extending heat equation into linear systems:

- (a) the heat diffusions along **multiple feature channels are correlated**
- (b) the diffusions along **horizontal and vertical directions** are not equivalent
- The **former** is straightforward as most of neural architectures output highly correlated features
- The **latter** is due to the **shape and texture anisotropy in visual objects which determines the heat diffusions along features**
- This is different with original heat equation which is spatial isotropy on a single channel.

❖ Heat equation in feature space

Uni-dimensional and observed variable into multi-dimensional and latent feature space

- Based on these **two guidelines**, firstly replace temperature u in the original heat equation with feature vector $z \in \mathbb{R}^C$ and use the steady-state condition $\frac{\partial z}{\partial t} = 0$ **to remove time dependence**, resulting in a Laplacian equation $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$.
- Extending Laplacian from spatial isotropy to anisotropy by adding a coefficient matrix S with size $C \times C$ as $\frac{\partial^2 z}{\partial x^2} + S \frac{\partial^2 z}{\partial y^2} = 0$.
- To allow self-prediction along horizontal and vertical directions, we decouple x and y axes in Laplacian into two first-order linear differential equations as:

First order linear:

- $\frac{\partial z}{\partial x} = A z, \frac{\partial z}{\partial y} = B z$, where A and B are invertible matrices with size $C \times C$ and $S = -A^2(B^2)^{-1}$

From continuous to discrete:

- continuous coordinates (x, y) by using discrete measure over $H \times W$ locations, converting continuous equation to difference over small segment (Δx or Δy) as follows:

$$\frac{\partial z}{\partial x} = A z, \frac{\partial z}{\partial y} = B z \quad \longrightarrow \quad \begin{aligned} z(x + \Delta x, y) - z(x, y) &= \Delta x A z(x, y) \\ z(x, y + \Delta y) - z(x, y) &= \Delta y B z(x, y) \end{aligned}$$

❖ QB-Heat

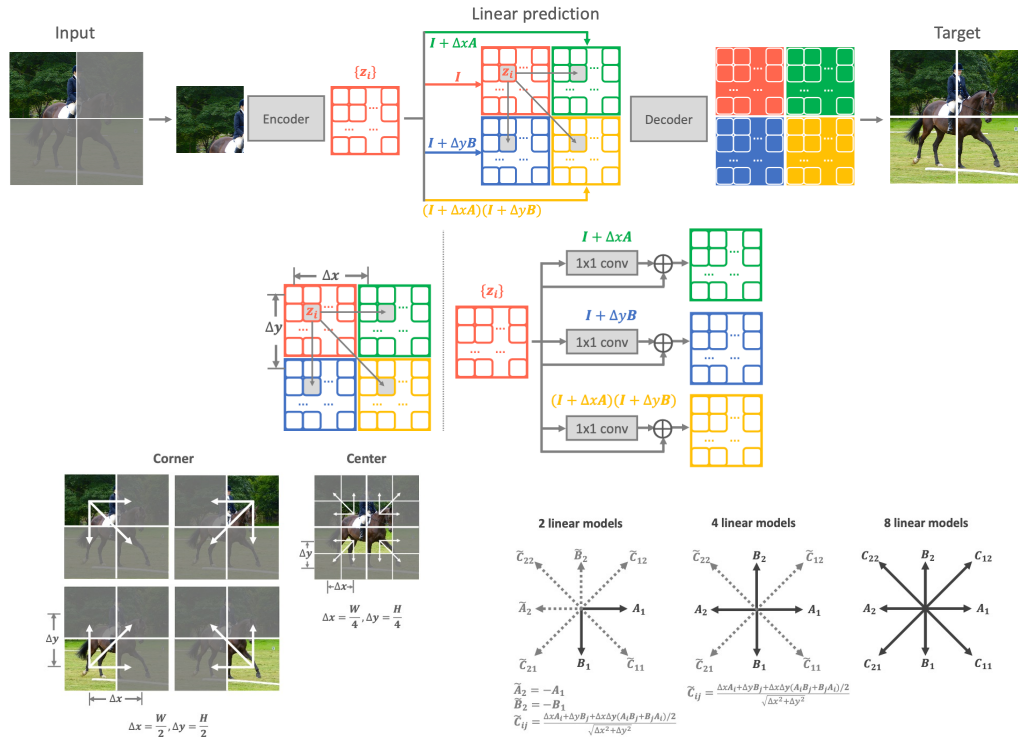
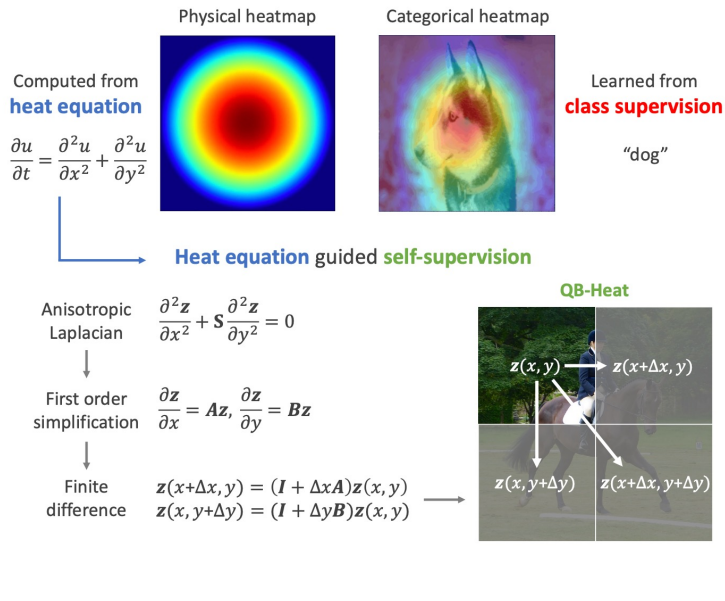


Figure 5. **Number of explicit linear models.** A solid arrow indicates an explicit linear model, while a dash arrow indicates an implicit model derived from the explicit counterparts.

❖ Architecture and Implementation

QB-Heat encoder:

- MobileFormer as encoder, which is a CNN based network (adding 6 global tokens in parallel to MobileNet).

QB-Heat predictor:

- The output features of the unmasked quarter-block are projected to 512 dimensions and followed by linear models to predict for masked blocks. This predictor is only used in pre-training and removed during inference.

QB-Heat decoder:

- Following MAE to apply a series of transformer blocks as decoder on both unmasked and masked quarter-blocks.

❖ Relation to MAE

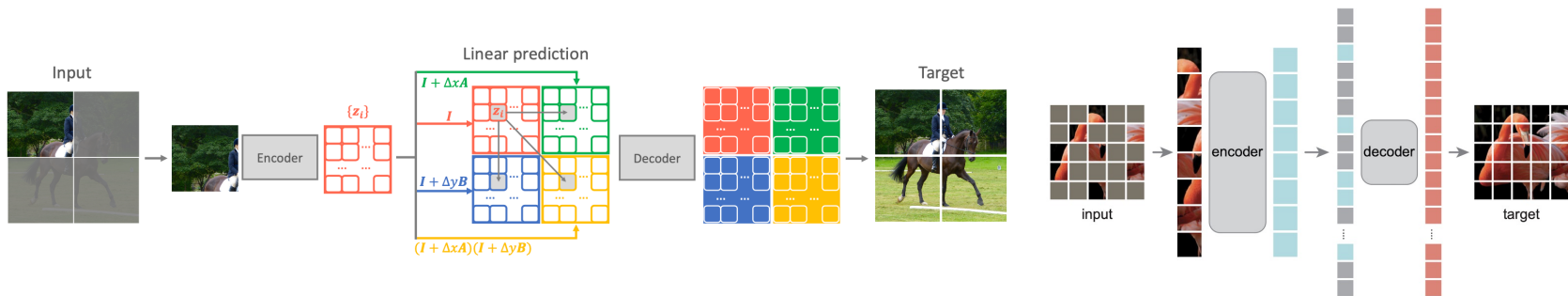
QB-Heat differentiates from MAE by **explicitly modeling feature derivatives using linear differential equations**, enabling **more regular masking** and **simpler prediction** to support more efficient CNN based networks.

More regular masking:

- Different with random unmasked patches in MAE, QB-Heat has a **single unmasked quarter-block**.

Simpler prediction:

- In QB-Heat, each masked patch is predicted from a **single unmasked patch with translation Δx or Δy** rather than aggregating all unmasked patches in MAE, thus resulting in **much lower complexity**.



Question?

❖ Evaluation: Decoder probing

Propose a new evaluation protocol for self-supervised pre-training

- Linear probing is sensitive to feature dimension and misses the opportunity of pursuing non-linear features

New protocol

- (a) can handle both **linear** and **non-linear** features
- (b) performs direct evaluation without fine-tuning
- (c) covers multiple visual tasks. It encourages exploration of pre-training a universal (or task-agnostic) encoder
- For each task, only the decoder is learnable while the pre-trained encoder (backbone) is frozen.

Classification decoders:

- (a) linear decoder (or linear probing) including global average pooling and a linear classifier
- (b) transformer decoder that adds a single transformer block before global pooling (denoted as tran-1)

Detection decoders:

- two DETR heads and one RetinaNet head.

❖ Evaluation: Decoder probing

encoder	madds	param	lin	encoder	madds	param	lin
Mob-v3 [29] [†]	217M	5.4M	36.3	Res-18 [27] [†]	1.8G	11.7M	52.5
Eff-b0 [42] [†]	390M	5.3M	42.2	Res-34 [27] [†]	3.6G	21.8M	57.4
Eff-b1 [42] [†]	700M	7.8M	50.7	MF-1.0G [12] [‡]	1.0G	13.5M	60.4
MF-285M [12] [‡]	285M	5.8M	51.6				

Table 1. **Linear probing results** of efficient networks pre-trained by MoCo-v2 [9]. “MF” (e.g. MF-285M) refers to Mobile-Former. [†] and [‡] indicate implementation in [21] and this paper respectively.

position (prediction offset)	lin	tran-1	ft	#models	lin	tran-1	ft
corner ($\Delta x=W/2, \Delta y=H/2$)	64.1	77.9	82.1	2	64.8	78.4	82.3
center ($\Delta x=W/4, \Delta y=H/4$)	64.2	77.9	82.4	4	65.0	78.5	82.4
corner + center	65.1	78.6	82.5	8	65.1	78.6	82.5

(a) **Position of the unmasked quarter-block.**

(b) **Number of linear models.**

Table 2. **QB-Heat ablation experiments** with Mobile-Former-3.7G on ImageNet-1K. We report top-1 accuracy (%) of two decoder probings, i.e. linear (lin) and transformer (tran-1), and fine-tuning with transformer decoder (ft). Two properties are observed: (a) multi-scale prediction (corner+center) is better than single scale, and (b) two explicit linear models ($\mathcal{A}$, $\mathcal{B}$ in Eq. (5)) are good enough. Default settings are marked in gray.

method	MF-285M	MF-1.0G	MF-3.7G
supervised	75.7	79.4	80.8
MoCo-v2	74.3	79.2	80.0
QB-Heat	75.8	80.5	82.5

Table 4. **Fine-tuning results**, evaluated on ImageNet-1K. QB-Heat consistently outperforms baselines over three models. The gain increases as the model gets bigger. tran-1 decoder is used for all methods.

pre-train	encoder	madds	param	fine-tune
MAE-Lite [45]	ViT-Tiny	1.2G	6M	76.1
QB-Heat	MF-285M	0.4G	6M	75.8
QB-Heat	MF-1.0G	1.4G	15M	80.5
iBOT [51]	ViT-S	4.6G	22M	82.3
MoCo-v3 [11]	ViT-S	4.6G	22M	81.4
MAE [25]	ViT-S	4.6G	22M	79.5
CMAE [30]	ViT-S	4.6G	22M	80.2
ConvMAE [22]	ConvViT-S	6.4G	22M	82.6
QB-Heat	MF-3.7G	5.5G	35M	82.5

Table 6. **Comparisons with previous results on ImageNet-1K.** All self-supervised methods are evaluated by end-to-end fine-tuning. All results are on an image size of 224.

$$\frac{E(\mathcal{A}^{\frac{1}{2}})}{E(\mathcal{B}^{\frac{1}{2}})} \approx \frac{E(\mathcal{A}^{\frac{1}{4}})}{E(\mathcal{B}^{\frac{1}{4}})}, \quad (6)$$

where the spectrum energy is computed as the sum of magnitude of eigenvalues as:

$$E(\mathcal{A}) = \sum_{k=1}^n |\lambda_k|, \quad E(\mathcal{B}) = \sum_{k=1}^n |\pi_k|, \quad (7)$$

epoch	$E(\mathcal{A}^{\frac{1}{2}})/E(\mathcal{B}^{\frac{1}{2}})$	$E(\mathcal{A}^{\frac{1}{4}})/E(\mathcal{B}^{\frac{1}{4}})$
200	0.9388	0.9418
400	0.8575	0.8559
800	0.8104	0.7997
1600	0.7336	0.7217
2400	0.6859	0.6813

Table 3. **Spectrum energy ratio** between horizontal and vertical matrices $\mathcal{A}$ and $\mathcal{B}$. Predictions at different scales ($\Delta x=W/4$, $\Delta y=H/4$ vs. $\Delta x=W/2$, $\Delta y=H/2$) have similar energy ratio, which becomes smaller as training schedule gets longer.

❖ Evaluation: Decoder probing

head			backbone											
model	madds	param	model	madds	param	pre-train	IN-ft	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L	
	(G)	(M)		(G)	(M)									
MF Dec 522	34.6	19.4	MF 3.7G	77.5	25.0	sup	–	40.5	58.5	43.3	21.1	43.4	56.8	
						moco2	✗	25.5 _(-15.0)	40.4	26.7	12.3	27.2	37.0	
						moco2	✓	31.7 _(-8.8)	48.3	33.5	16.1	33.4	45.5	
						QB-Heat	✗	43.5 _(+3.0)	61.3	47.2	23.2	47.1	60.6	
						QB-Heat	✓	45.5 _(+5.0)	64.0	49.3	25.2	49.1	63.5	
						sup	–	38.3	56.0	40.8	19.0	40.9	54.3	
	32.3	18.6	MF 1.0G	20.4	11.7	moco2	✗	30.3 _(-8.0)	46.0	32.3	15.1	32.1	42.5	
						moco2	✓	39.0 _(+0.7)	56.8	41.8	19.2	41.8	55.3	
						QB-Heat	✗	42.6 _(+4.3)	60.4	46.2	22.7	46.3	59.9	
						QB-Heat	✓	44.0 _(+5.7)	62.5	47.2	23.5	47.6	61.1	
						sup	–	35.2	52.1	37.6	16.9	37.2	51.7	
						moco2	✗	31.8 _(-3.4)	47.8	34.1	14.9	33.3	45.6	
31.1	18.2	MF 285M	5.6	4.9	moco2	✓	39.9 _(+4.7)	57.9	42.7	19.0	43.1	57.1		
					moco2	✓	39.7 _(+4.5)	57.6	42.7	20.4	42.6	56.4		
					QB-Heat	✗	41.6 _(+6.4)	59.2	45.0	21.4	45.2	58.4		
					sup	–	34.1	51.3	36.1	15.5	36.8	50.0		
					moco2	✗	12.2 _(-21.9)	24.1	10.7	5.3	13.0	19.3		
					moco2	✓	19.1 _(-15.0)	33.1	18.4	8.6	19.6	29.3		
MF Dec 211	15.7	9.2	MF 3.7G	77.5	25.0	QB-Heat	✗	36.7 _(+2.6)	53.8	39.5	17.2	39.6	53.5	
						QB-Heat	✓	41.0 _(+6.9)	59.3	44.2	20.9	44.5	58.2	
						sup	–	31.2	47.8	32.8	13.7	32.9	46.9	
						moco2	✗	16.9 _(-14.3)	29.7	16.4	7.7	17.6	25.8	
						moco2	✓	30.6 _(-0.6)	46.7	32.1	14.4	32.0	45.2	
						QB-Heat	✗	35.7 _(+4.5)	52.5	38.5	16.9	38.6	51.5	
	13.4	8.4	MF 1.0G	20.4	11.7	QB-Heat	✓	39.3 _(+8.1)	56.8	42.0	18.9	43.1	56.3	
						sup	–	27.8	43.4	28.9	11.3	29.1	41.6	
						moco2	✗	22.1 _(-5.7)	35.7	22.8	9.6	22.4	34.4	
						moco2	✓	32.7 _(+4.9)	49.0	34.6	14.5	35.1	48.8	
						QB-Heat	✗	33.0 _(+5.2)	49.3	35.1	15.6	35.2	48.5	
						QB-Heat	✓	35.8 _(+8.0)	52.8	38.3	16.5	38.4	51.5	

Table 5. COCO object detection results on val2017 for *frozen* backbone pre-trained on ImageNet-1K. Evaluation is conducted over three backbones and two heads that use MobileFormer [12] end-to-end in DETR [5] framework. Our QB-Heat significantly outperforms MoCo-v2 and supervised baselines. Fine-tuning on ImageNet-1K provides consistent improvement. Initial “MF” (e.g. MF-Dec-522) refers to MobileFormer. “IN-ft” indicates fine-tuning on ImageNet-1K. MAdds is based on the image size 800×1333.

model	query	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L	madds (G)	param (M)
DETR-DC5 [5]	100	43.3	63.1	45.9	22.5	47.3	61.1	187	41
Deform-DETR [52]	300	46.2	65.2	50.0	28.8	49.2	61.7	173	40
DAB-DETR [38]	900	46.9	66.0	50.8	30.1	50.4	62.5	195	48
DN-DETR [33]	900	48.6	67.4	52.7	31.0	52.0	63.7	195	48
DINO [49]	900	50.9	69.0	55.3	34.6	54.1	64.6	279	47
QB-MF-DETR	100	49.0	67.8	53.4	30.0	52.8	65.8	112	44

Table 7. Comparisons with DETR based models on COCO. QB-MF-DETR uses MobileFormer (MF-3.7G) as backbone, which has similar FLOPs and model size with ResNet-50 used in other methods. MAdds is based on image size 800×1333.

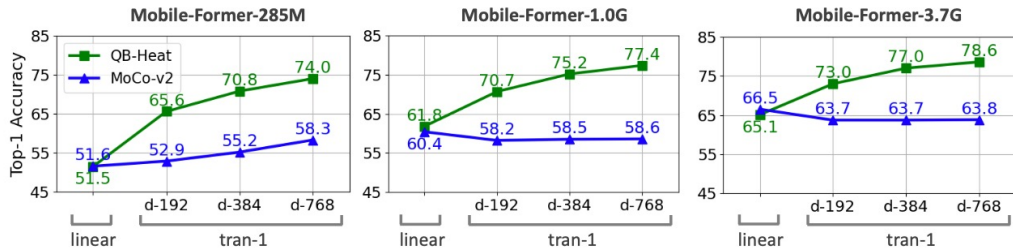


Figure 8. Decoder probing on ImageNet-1K. QB-Heat is on par with MoCo-v2 on linear probing, but outperforms on tran-1 probing. The gain increases as the decoder gets wider.

❖ Connection with diffusion model & Limitation

- Diffusion models study noise diffusion along **the path from image to noise**.
- QB-Heat studies a different type of diffusion: **the semantic heat diffusion of feature vector across 2D space**.

But they essentially share a common insight:

- Learning diffusion rate as a function of signal. Specifically, in diffusion model, the noise ϵ_t at step t is a function of x_t .
- In contrast, QB-Heat models the feature change using linear equations as $\frac{\partial z}{\partial x} = \mathbf{A}z, \frac{\partial z}{\partial y} = \mathbf{B}z$.

QB-Heat has a major limitation:

- not working well for vision transformers (ViT).
- Specifically, QB-Heat does not have a chance to see tokens beyond a quarter-block in pre-training, but all tokens of an entire image are used during inference.
- This discrepancy becomes more critical for transformers than CNN, as long range interaction is directly modeled via attention in transformer.

❖ New self-supervised learning guided by heat equation

- Extension heat equation from **a single measurable variable to high dimensional latent feature vector**, and simplifies it into first order linear differential equations.
- Based on such simplification, we develop a new masked image modeling that learns **to linearly predict three masked quarter-blocks** from a single unmasked quarter-block.
- QB-Heat not only enables masked image modeling for efficient CNN based architectures, but also provides **strong task-agnostic encoders** on both image classification and object detection.

Question?

Thank you for attention.